Carry out the investigation and collect data

All four setups run together. Take measurements every 10 minutes of simulated time.

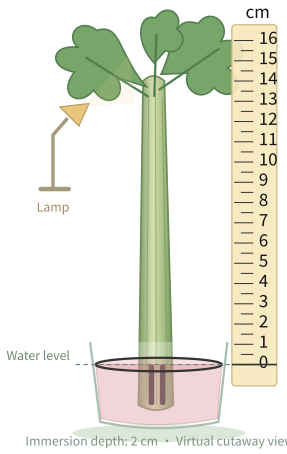
✦Conditions kept the same: Temperature of 25°C, the same humidity (濕度) and air flow, the same red water concentration, a depth of 2 cm in the water, and leaf stalks of similar length and thickness. Setups A, B and C have the same total leaf area. Setup D has no leaves.

In a real experiment: Keep the experiment and timer running while you measure. Read and record the data quickly at each planned time. Do not stop the timer to take measurements. This simulation pauses the picture to help you observe and record. The time you spend doing this is not part of the 30 simulated minutes.

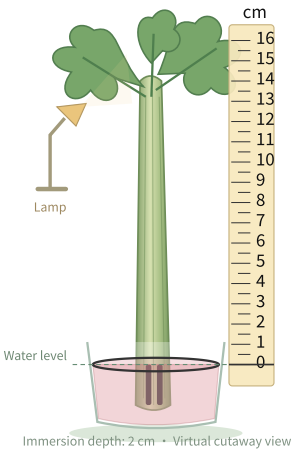
Read the rulers in the static images and record your own readings. These images use the same teaching model as the current VL.

0 min

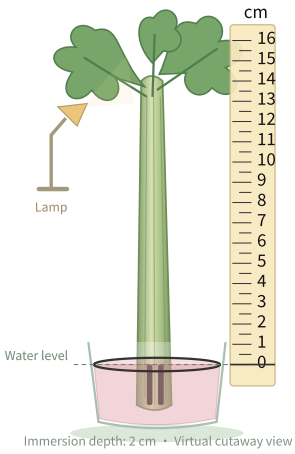
Setup A • With leaves
10 cm • 1200 lux



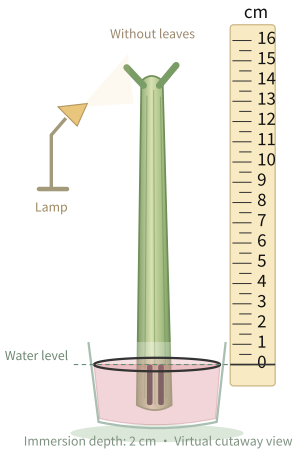
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Your measurement record

Simulation time (min)	Distance moved upwards by the red water trace (cm)			
min	Setup A: light distance 10 cm With leaves	Setup B: light distance 20 cm With leaves	Setup C: light distance 30 cm With leaves	Setup D: light distance 20 cm Without leaves
0				
10				
20				
30				

After entering a reading, press Enter to move to the next box. After Setup D, it moves to the confirm button. Your first readings are kept, and changes are also recorded. The platform does not replace your readings with reference values.

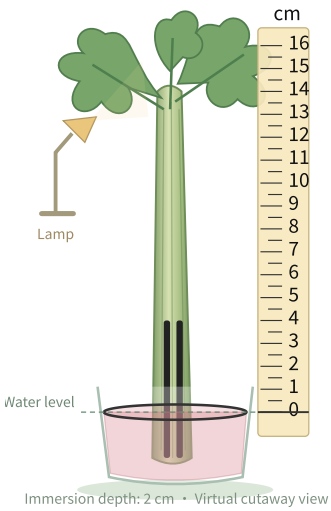
If you revise an answer, keep the first answer and write the change on the two lines below.

Carry out the investigation and collect data

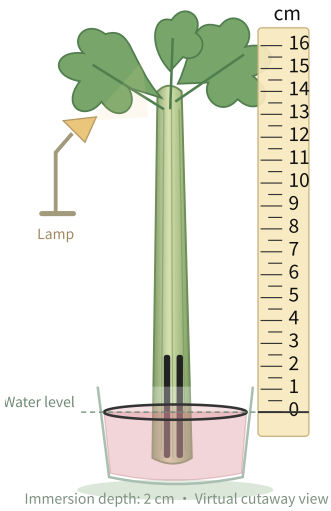
Read the rulers in the static images and record your own readings. These images use the same teaching model as the current VL.

10 min

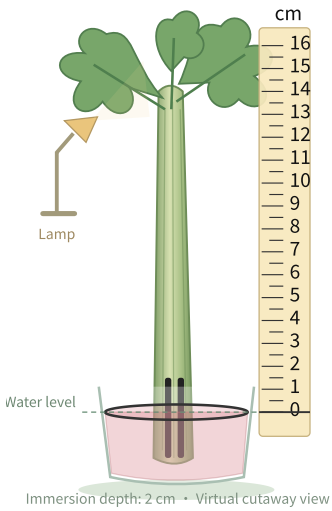
Setup A • With leaves
10 cm • 1200 lux



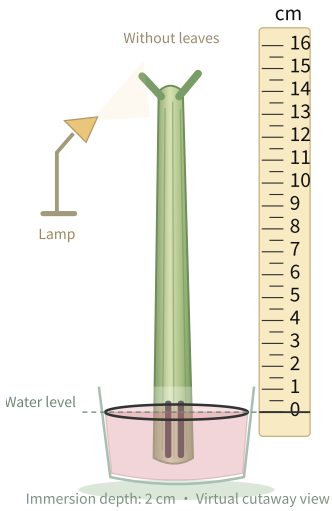
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux

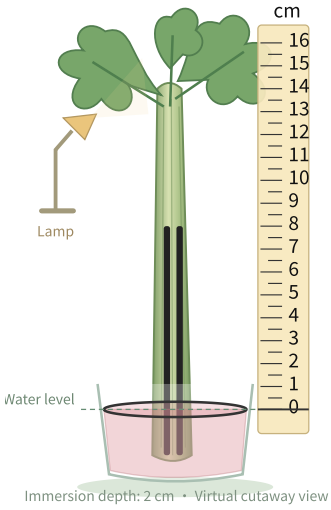


Setup D • Without leaves
20 cm • 300 lux

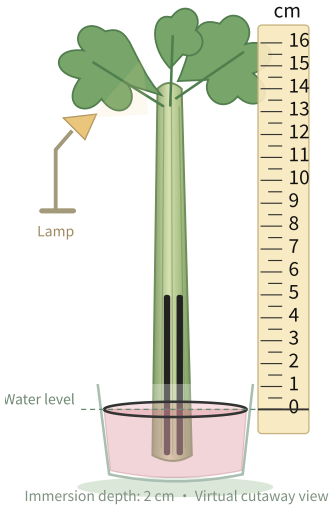


20 min

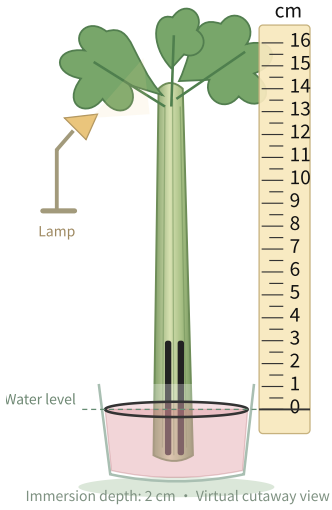
Setup A • With leaves
10 cm • 1200 lux



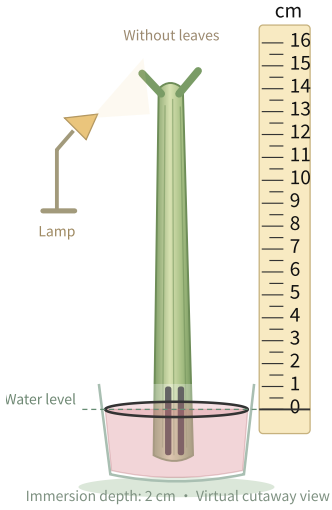
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux

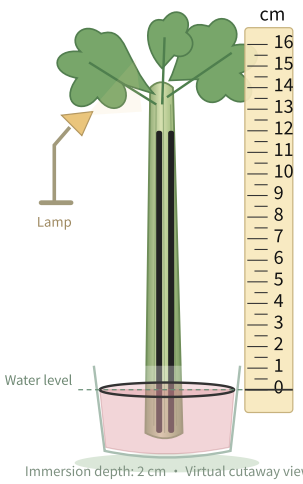


Drag the ruler left or right to place it beside the leaf stalk. Its 0 mark stays at the water surface, so you cannot move it up or down. Read the height of the top of the red water trace. Enter the reading to one decimal place. The arrows beside each reading change it by 0.5 cm.

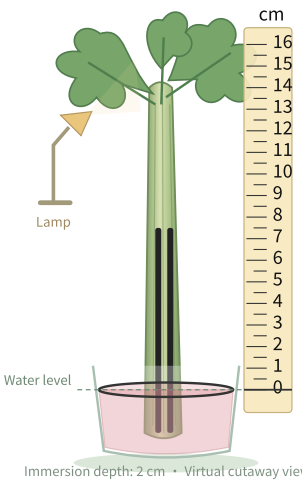
Each 10 minutes of simulated time takes about 3 seconds. This is a model for learning. It does not predict the exact results of a real experiment.

30 min

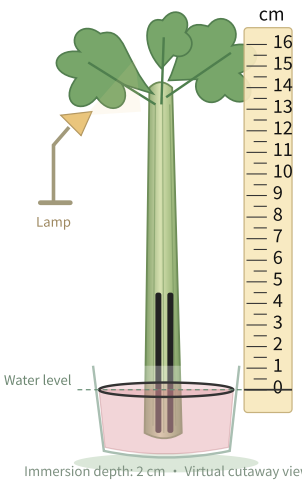
Setup A • With leaves
10 cm • 1200 lux



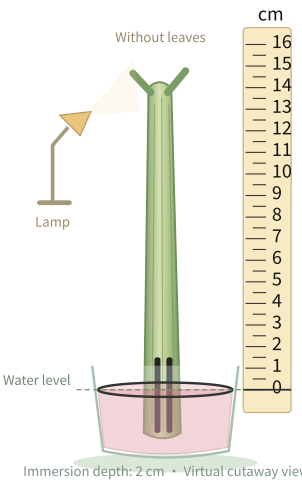
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Calculate the average upward speed

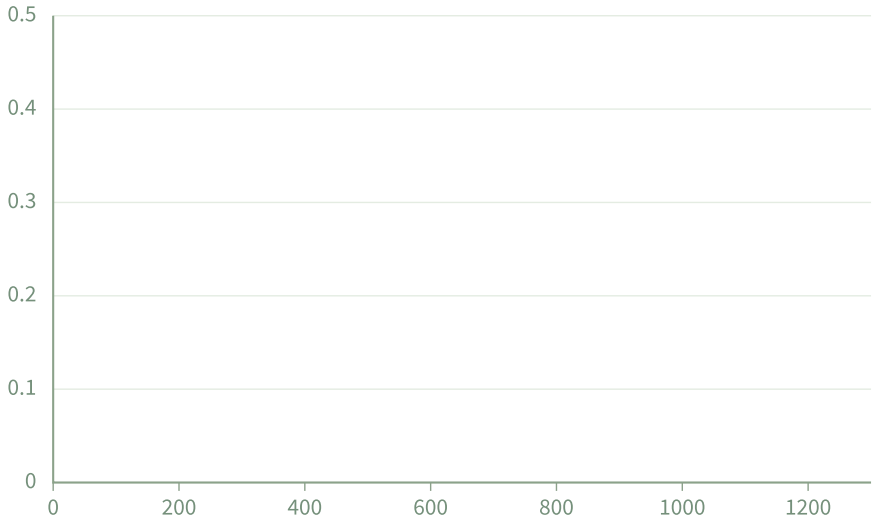
Use your readings and the time taken to calculate the average upward speed for each setup. Record each answer to two decimal places.

Average upward speed (cm/min) = distance moved upwards by the red water trace (cm) ÷ time taken (min)
Distance moved upwards = reading at 30 minutes − reading at 0 minutes. The time taken in this investigation is 30 minutes.

Setup	Average upward speed (cm/min)
Setup A • With leaves • 10 cm	
Setup B • With leaves • 20 cm	
Setup C • With leaves • 30 cm	
Setup D • Without leaves • 20 cm	

Light intensity (光強度) and the upward speed of the red water trace

Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

With leaves: Setups A, B and C | Without leaves: Setup D (a separate comparison point)

Move the pointer over the graph to see its (X, Y) coordinates (坐標). Choose a setup, then click the graph to plot its point. Click an existing point to edit its coordinates (坐標) in the boxes below. Select ‘Confirm / update point’ to save. You can also type coordinates (坐標) to plot a point. Record Y coordinates (坐標) to two decimal places. Join the points for setups with leaves with a smooth curve. Show Setup D separately.

Use your graph to reach a conclusion

Circle one answer in each set of brackets in the sentence.

1. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will （increase ／ decrease ／ show no clear change ／ show no consistent trend）.

2. What is the role of leaves?

At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is （faster ／ slower ／ similar）.

This comparison supports the idea that leaves （help water move upwards ／ stop water moving upwards ／ have no effect on water moving upwards）.

3. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?（The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). ／ The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). ／ If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).）

If you revise an answer, keep the first answer and write the change on the two lines below.

Give out C only after the original investigation answers are fixed.